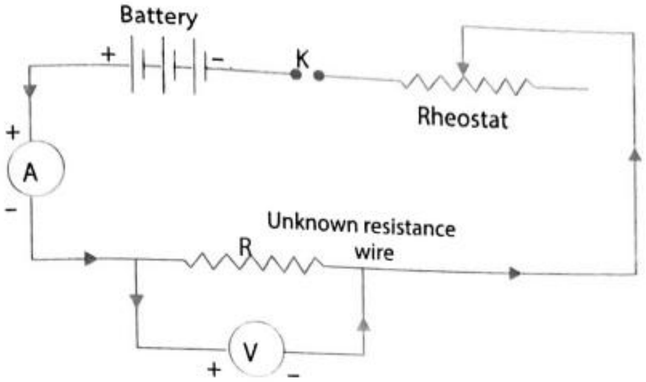
ACTIVITY – CORRECTION OF AN ELECTRIC CIRCUIT

AIM: To draw the diagram of a given open circuit comprising at least a battery, resistor/rheostat, key, ammeter and voltmeter. Mark the components that are not connected in proper order and correct the circuit and also the circuit diagram.

APPARATUS AND MATERIALS: A battery eliminator or a battery (0 to 6 V), rheostat, two resistors, one-way key, D.C. ammeter (0-3) A and a D.C. voltmeter (0-3) V.

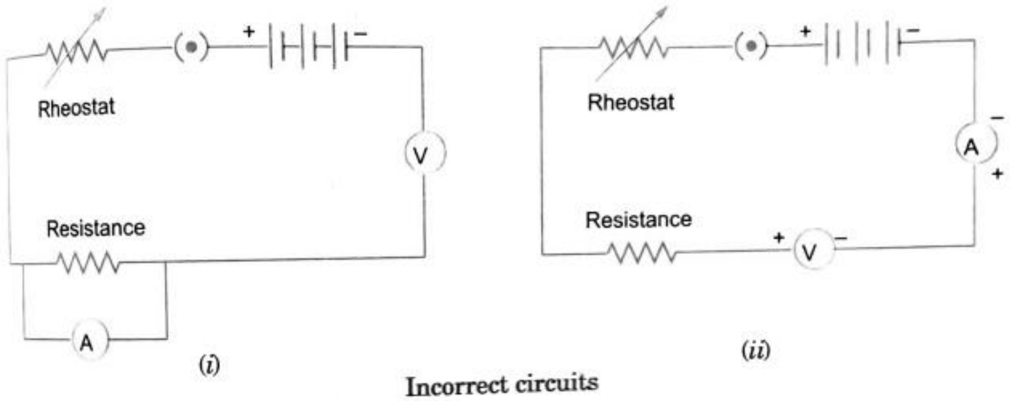
THEORY: An open circuit is the combination of primary components of electric circuit in such a manner that on closing the circuit, no current is drawn from the battery.



If the circuit is not working, it can have many causes. Some of them are:

(i) Broken or fuse wires. (ii) Loose connections.

(iii) Exhausted battery. (iv) Improper connection of components



Incorrect circuits

PROCEDURE:

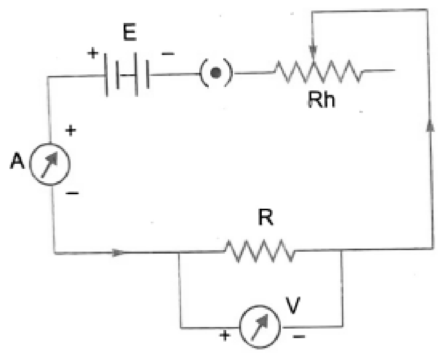
Ammeter: It should be connected in series, with the battery eliminator.

Voltmeter: It should be connected in parallel to the resistor.

Rheostat: It should be connected in series with the battery eliminator.

Resistance coil: It should be connected in series with the circuit.

One-way key: It should be connected in series to the battery eliminator.



Corrected circuit diagram

PRECAUTIONS:

1. Check which components are connected in proper order and which are not.
2. Take out the key plug before connecting the components.